Exercise list R9 — Mergers and acquisitions

Answer key (full derivations in `lista-r9-mergers-acquisitions-solutions.pdf`)

1. True or false: valuation and acquisitions (Higgins 9.1)

(a) False: targets' shareholders gain (20%+ at announcement); acquirers' gains are doubtful. **(b)** False: discount the target's FCF at the *target's* cost of capital (risk of the cash flows). **(c)** False: the premium pays for improvements; a well-run firm has little to improve (e.g. 5% vs 40%). **(d)** False: liquidation needs someone with control willing to liquidate. **(e)** True: price \$30 vs insiders' \$80 makes an MBO attractive (and creates a conflict of interest). Matches p. 425–426.

2. LBO candidates and MBOs (G&T 20.5, 20.8)

20.5 Stable cash flows carry more riskless debt (worst case 90 vs 40 for the same mean 100), and generic products/redeployable assets keep distress costs low; so the tax and free-cash-flow benefits of leverage come cheap. **20.8** The problem is the managers' small stake and discretion, not their identity: ownership rises (5% → 15% of each \$1 created, ×3) and debt service commits free cash flow.

3. Target prices around announcements (Higgins 5.10)

(a) No: a predictable response to public news violates semistrong efficiency. **(b)** Buy on announcement, sell after two days: with one third of the 20-point jump on day 0, $128.5/115.17 - 1 = 11.6\%$. **(c)** The response becomes instantaneous. **(d)** Exploiting the inefficiency removes it. **(e)** No: the price is an expectation that includes a higher (rival) bid, e.g. $0.3(75) + 0.6(60) + 0.1(35) = 62 > 60$.

4. Sources of LBO premiums (Amaro de Matos 7.2.4 #3)

Tax shields ($\tau D = 0.34 \times 600 = 204$ in an illustration); lower agency costs (managerial ownership, debt disciplines free cash flow); transfers from bondholders, preferred stockholders and employees (small); managers' private information; efficiency of private ownership. Matches p. 271.

5. Grossman–Hart with a toehold (Amaro de Matos 8.1.5 #1)

Shareholders tender only if $p \geq v$, so with no toehold $\Pi_r = v - p - c < 0$. With toehold θ: $\Pi_r = v - p(1-\theta) - c > 0 \iff \theta > 1 + (c-v)/p$ (at $p = v$: $\theta > c/v$). Net of the toehold's initial value θq: $\theta(v-q) > c + (1-\theta)(p-v)$. Example $v = 100$, $q = 80$, $c = 1$, $\theta = 0.1$, $p = 100$: $\Pi_r = 9$, net gain 1. Matches p. 271.

6. News Corp. and Dow Jones (Higgins 9.3)

(a) $60 \times 82\text{m} + 1.46\text{bn} = 4.92 + 1.46 = \$6.38$ billion (equity plus assumed debt). **(b)** $(60-33) \times 82\text{m} = \$2.21$ billion, an 81.8% premium. Matches p. 426.

7. Procureps (Higgins 9.9)

(a) $MV_P = 60$ ($p = \$60$), $MV_{V1} = 8$, $MV_{V2} = 35$; EPS = 2 caps shares at $3/2 = 1.5$m, so 0.5m new shares worth \$30m: max premium $30/8 - 1 = 275\%$ for V1, $30/35 - 1 = -14\%$ for V2. **(b)** The rule favours low-P/E targets and blocks high-growth ones; it is unrelated to value (buying V2 at market

dilutes EPS to \$1.89 but keeps the price at \$60; buying V1 at 275% keeps EPS at \$2 but drops the price to \$45.33). Matches p. 427.

8. Hansen, uniform case (Amaro de Matos 8.2.6 #3)

Cash C is accepted iff $v \le C$; $E(w \mid C) = 0.75C$, so $E(\Pi_r \mid C) = (C/100)(0.75C - C) < 0$ for all C. Stock β: all types accept iff $\beta \ge 100/(x+150)$ (worst type $v = 100$, since $v/(x+1.5v)$ rises); acquirer gains iff $(1-\beta)(x+75) \ge x \iff \beta \le 75/(x+75)$. Compatible iff $25x \le 3750 \iff x \le 150$ (e.g. $x = 100$: $\beta \in [0.4, 0.4286]$, gain 5 at $\beta = 0.4$); for $x \ge 150$ no cutoff v^* works either. Matches p. 276.

9. Fishman (1988), partial revelation (Amaro de Matos 8.2.6 #1)

All $v_1 \ge r$ bid $\bar{p}(r)$, all $v_1 < r$ bid v: bidder 2 learns only the interval, because its entry decision depends only on whether $\mathbb{E}[g(v_1) \mid \text{message}] \ge c_2$. With v_1 observable bidder 1 bids v and bidder 2 enters iff $v_1 < w$, $g(w) = c_2$, with $w > r$; types in $[r, w)$ are deterred only under signalling. Example $U[0,1]$, $v = 0.6$, $c_2 = 0.02$: $r = 0.654$, $w = 0.8$, $\bar{p}(r) = 0.62$. Matches p. 276.

10. Jegadeesh–Chowdhry, no pooling (Amaro de Matos 8.1.5 #8)

$Z_M = 0.4(40) + 0.6(80) = 64$; $G(64, 0.3; 80) = 27.2 \ln 64 = 113.12$; $G(64, 0.3; 40) = 7.2 \ln 64 = 29.94$. Low type deviates to bid 40 with α' from $\ln 40\,(20 + 40\alpha') = 113.12$: $\alpha' = 0.2666$; then $G(40, \alpha'; 40) = 39.34 > 29.94$, and the high type is indifferent. No pooling; a smaller toehold signals a lower valuation. Matches pp. 274–275 (book: 39.337 with rounded α').

11. Flip-over pill with voting new shares (Tirole 11.1)

With m free voting shares per untendered share, the raider keeps control iff $\kappa' \ge (1+m)/(2+m)$ (2/3 for $m = 1$). Holdouts gain $\Delta_u = m\hat{v}/2$, raider shares lose $\Delta_r = m\hat{v}/[2(1+m)]$, with $\kappa'\Delta_r = (1-\kappa')\Delta_u$. Then $\beta = P/(1+\Delta_u)$, $\pi = \beta(\hat{w} - \Delta_u)$, bid $P = 1 + \Delta_u$ if $\hat{w} > \Delta_u$, optimal pill $\Delta_u^* = \hat{w}$: Tirole's results with $\Delta \to \Delta_u$. Example $v = 2$, $\hat{v} = 3$, $\hat{w} = 2$, $m = 1$: $P = 2.5$, $\pi = 0.5$ (non-voting: $P = 2$, $\pi = 1$). No book answer.

12. Hansen, stock dominates cash (Amaro de Matos 8.2.6 #4)

Take $\beta^* = C/[x + w(C)]$. Hansen's $w' < (x+w)/v$ makes $v/[x+w(v)]$ increasing, so β^* is accepted iff $v \le C$ (same $F(C)$). Then $W^{\text{stock}} - W^{\text{cash}} = F(C)\{C - \beta^*[E(w \mid C) + x]\} = F(C)\beta^*[w(C) - E(w \mid C)] \ge 0$. Example $U[0, 100]$, $w = 1.2v + 30$, $x = 200$, $C = 60$: cash gain 3.6, stock gain 7.89. Book refers to Hansen (1987), p. 80.

13. Fishman (1989), cash vs debt face value (Amaro de Matos 8.2.6 #8)

$p_c - p_d = (1-\gamma)[v_1^- - p_d] < 0$ since $p_d = \mathbb{E}\min[\max(v, \tilde{v}_2^+), v_1^+] \ge v > v_1^-$. Cash is paid in the bad state too, so it is a costlier signal per unit of face value and a lower face value suffices. Example $\gamma = 0.6$: $p_d = 0.675$, $p_c = 0.565$. Matches pp. 277–278.

14. Shleifer–Vishny with arbitrary P (Tirole 14.2)

$U_{bi} = \hat{\alpha} I_i + \hat{\kappa} I_j$, $\hat{\alpha} = \alpha - (1-x)(1-\nu)(\rho_0 - P)$, $\hat{\kappa} = \kappa + x(1-\mu)(\rho_0 - P)$, $\hat{\alpha} + \hat{\kappa} = \alpha + \kappa$. Capacity $I_i = [A_i + x(1-\mu)(\rho_0 - P)I_j]/[1 - x\rho_0 - (1-x)(1-\nu)P]$ (strategic complements if $P < \rho_0$);

symmetric $I = A/\{1-\rho_0[x+(1-x)(1-\nu)]\}$, independent of P. Example ($x = 0.8$, $\nu = 0.3$, $\rho_1 = 1.5$, $\rho_0 = 0.6$, $P = 0.3$): $\hat{\alpha} = 0.242$, $\hat{\kappa} = 0.168$, $I = 2.294A$. Matches p. 618.

15. Liquidity management and acquisitions (Tirole 14.3)

(i) $U = (\rho_1 - 1)I + (1-x)J\int_0^{\rho^*}[\rho_1 - P - \rho]dF$, breakeven $\rho_0 I + (1-x)J\int_0^{\rho^*}[\rho_0 - P - \rho]dF = I - A$; the FOC gives $(\rho_1 - s)(1-\rho_0) = (\rho_1 - 1)(s - \rho_0)$, so $s = P + \rho^* = 1$ and $\rho_0 + L^* = 1$. **(ii)** Seller: $\max_P PF(\rho_0 + L - P)$; $P = F(1-P)/f(1-P)$, unique in $(0,1)$ by the hazard-rate condition; $dP/dL = G'/(1+G') \in (0,1)$, $G = F/f$. Exponential $\lambda = 2$: $P = 0.604$, $dP/dL = 0.688$. **(iii)** Spike at $\bar{\rho}$: $P \to 1 - \bar{\rho}$, $F(\rho^*) \to 1$ (uniform $\pm\varepsilon$: $P = 1 - \bar{\rho} - \varepsilon$). Matches pp. 618–619.